Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/14/2004
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	Effect of Spaceflight on Microbial Gene Expression and Virulence		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	09/01/2002	**End Date:**	11/30/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hammond, Timothy (Tulane University Health Sciences Center/Veterans Affairs Medical Center) Buchanan, Kent (Tulane University Health Sciences Center) Schurr, Michael (Tulane University Health Sciences Center) Pierson, Duane (NASA Johnson Space Center)		
Grant/Contract No.:	NCC2-1362		
Performance Goal No.:			
Performance Goal Text:			

Human presence in space, whether permanent or transient, will be accompanied by the presence of microbes. The changes that occur to microorganisms during space flight are not fully understood, and the potential of a negative impact upon health, safety, and performance of crewmembers requires further investigation. Space flight studies have demonstrated changes in bacterial and fungal growth, morphology, and metabolism, and ground-based studies have indicated changes in bacterial virulence characteristics and protein production during growth in modeled microgravity, a laboratory environment that stimulates some aspects of spaceflight. Thus, during long-duration space flight, the virulence of pathogenic microbes may be enhanced and normally harmless commensal or environmental microbes may become pathogenic. This effect would be intensified by several factors, including: 1) the increased duration of missions, 2) a greater microbial diversity resulting from the use of regenerative life support systems, and 3) a potentially compromised human immune system during flight. Investigation into the effects of space flight on microbial characteristics is difficult as flight experiments are expensive and opportunities are rare. Therefore, ground-based analog systems have been developed for cellular and microbial testing; the most popular system being NASA designed rotating wall vessel (RWV) bioreactors, which simulate many of the mechanistic effects of microgravity on biological systems. The objective of this research is to compare microbial gene expression and virulence attributes in microorganisms during extended space flight to the same organisms cultured under normal gravity and in the modeled microgravity of the RWV. The latter comparison is essential to verify the applicability of the RWV bioreactors to simulate in-flight conditions and to use this system to access the probability of in-flight infectious disease. This proposal is to fly two copies of the automated single locker MOBIAS (Multiple Orbital Bioreactor with Instrumentation and Automated Sampling) culture unit, to allow simultaneous analysis in multiple replicates of four difference bacterial and fungi. This study develops a foundation from which future in-flight research can be based and which may also hold commercial promise for translation to the clinical bedside. The objective of this research is to determine changes in microbial gene expression and virulence attributes during extended spaceflight. These results will be compared to those obtained when the same experiments are performed with the identical microbes cultured under ground-based conditions of modeled microgravity (MMG) in NASA-designed bioreactors. This comparison is vital in determining the value of the NASA-designed bioreactors in suspension culture of microbes under MMG for use in: (a) simulating the probability of in-flight infectious disease; and, (b) developing commercial potential for the bioreactors for translation to the clinical bedside. Four model microbes, two bacterial and two fungal, are proposed for use in these studies: a). the obligate pathogen Salmonella typhimurium, b). the opportunistic pathogen, Pseudomonas aeruginosa, c). the commensal organism and opportunistic pathogen, Candida albicans; and, d). the classic model eukaryotic microbe, Saccharomyces cerevisiae. These organisms were shown to be gravity sensitive in earlier flight experiments and/or ground-based simulations. The objective of the proposal will be met by achieving the following Specific Aims:
Specific Aim 1). Whole genome microarray-mediated gene expression profiling of S. typhimurium, P. aeruginosa, C. albicans, and S. cerevisiae, in response to extended spaceflight and during suspension culture in MMG, as compared to normal gravity controls.

Specific Aim 2). Determination of the effect of extended spaceflight on the virulence potential of the pathogenic microorganisms immediately following their return from spaceflight.

Task Description:

This study will be the first of its kind to investigate comprehensive changes in microbial gene expression which occur during extended spaceflight and their comparison to the expression profiles of the same microbes grown as suspension cultures under MMG in the NASA-designed bioreactors. Based upon our published studies using MMG culture of two of the microbes listed above, S. typhimurium and S. cerevisiae, we anticipate that results generated during the course of the proposed studies will be instrumental to understanding the effect of spaceflight on microbial gene expression, pathogenicity, and the risk of in-flight infectious disease. Moreover, these studies will provide a solid foundation for testing the commercial applicability of ground-based microbial suspension culture under MMG for the development of vaccines and other novel countermeasures, which are not achievable by any other ground-based means, for the treatment and prevention of infectious diseases occurring during spaceflight and on Earth. This study will be the first of its kind to investigate comprehensive changes in microbial gene expression and virulence parameters which occur during extended spaceflight and their comparison to the expression profiles observed when the same microbes are grown under normal gravity and as suspension cultures under MMG. Accordingly, the proposed studies will be instrumental to gain a better understanding of the effect of spaceflight on microbial gene expression, pathogenicity, and the risk of in-flight infectious disease. Moreover, these studies are essential to verify the applicability of the RWV bioreactors to simulate in-flight conditions and to use this ground-based system to assess the probability of in-flight infectious disease. This study develops a foundation from which future in-flight research can be based and may also hold commercial promise for translation to the clinical bedside via the development of vaccines and other novel countermeasures for the treatment and prevention of infectious diseases occurring during spaceflight and on Earth. The proposed study will address two key questions that will benefit the NASA HEDS program, subsequent flight experiments, and ground-based vaccine development for a variety of diseases. First, this study will focus on the comparison of global changes in microbial gene expression following growth in the ground-based, RWV bioreactors (i.e. MMG environment) and in the true microgravity environment encountered during spaceflight. Without these critical comparisons to serve as controls for an initial baseline, and thus establish the scientific basis for the validity of comparison between spaceflight and suspension culture in MMG bioreactors, ground-based research results will continue to suffer from the stigma that these observations are not comparable to activity in space. Second, this study will compare putative changes in microbial virulence observed in the MMG bioreactor and that may be occurring during flight. This information will not only serve as the foundation for future flight experiments throughout the scientific community, but also as the initial template for discovery of countermeasures against microbial infection during flight. This proposal is responsive to the NRA 01-OBPR-03 under the Fundamental Space Biology Division of OBPR, specifically focusing on the Cellular and Molecular Biology research element. This proposal is consistent with flight experiments using simple organisms that are "biological systems known or believed to be influenced by gravitational forces." Moreover, the research outlined in this proposal directly addresses OBPR Fundamental Space Biology Division Goals to "Use microgravity and the other characteristics of the space environment effectively to enhance our understanding of fundamental biological processes" and "Develop the foundation of fundamental biological knowledge required to enable a long-duration human presence in space." This proposal achieves these goals by investigating the NASA Critical Path Roadmap questions:

11b (Priority 1): Does spaceflight alter microbial growth rates, mutation rates or pathogenicity? 11c (Priority 1): Does the spacecraft environment exert a selective pressure on environmental microorganisms which presents the crew with increased health risks? 7.19 (Priority 1): Does spaceflight alter host-microbe balance?

38a, 39a (Priority 1): What technologies can be used to detect slow, gradual changes in the chemical and microbial environment (work with ALS and Environmental Health)?

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will develop a foundation from which future ground-based research can be based and also holds commercial promise for translation to the clinical bedside. These investigations will be the first of their kind to investigate comprehensive changes in microbial gene expression which may occur during extended spaceflight and their comparison to the expression profiles of the same microbes grown as ground-based suspension cultures under MMG in the NASA-designed bioreactors. The latter comparison is essential to verify the applicability of the MMG bioreactors to simulate in-flight conditions and to use this system to assess the probability of in-flight infectious disease. However, these studies will also provide a solid foundation for testing the commercial applicability of ground-based microbial suspension culture under MMG for the development of vaccines and other novel countermeasures, which are not achievable by any other ground-based means, for the treatment and prevention of infectious diseases occurring on Earth. For example, the PI has previously demonstrated that when cultivated under MMG, the enteric pathogen, S. typhimurium, is more potent in terms of its disease-causing potential. Expanding this work to include other important pathogens could provide significant insight about the microbial response to MMG, and could provide clues for the functioning of known virulence systems, or the identification of novel, uncharacterized microbial virulence strategies, which could lead directly to the development of countermeasure for infectious human disease.
Task Progress:	Task Progress for current year: Long-term microbial survival profiles (100+ days) have been determined for each of the four organisms to be used in these studies. We have also determined optimal media composition and temperature on the long-term viability and robustness of each of the four organisms. RNA integrity profiles for each of the four organisms following storage in fixative at various temperatures has been determined. Due to the temporary grounding of the Shuttle fleet following the Columbia accident, a portion of this science, Yeast GAP1 (involving the yeast S. cerevisiae) was successfully flown to the International Space Station (ISS) on board a Russian Progress Rocket - flight mission 13-P in January 2004. The experiment went flawlessly and we are awaiting the return of the Yeast-GAP experimental payload. Evidence that the Yeast GAP experiment was successfully performed on board ISS are as follows: First, the real time streaming video, and still images which astronaut Michael Foale shot during experiment activation and termination show that the yeast samples on the International Space Station grew in every chamber of the flight hardware as expected, as the growth media changed as expected from clear yellow to white, and the gap exchange membranes moved consistent with gas production. Second, parallel ground controls, prepared contemporaneously with the flight samples, and run almost simultaneously, show excellent growth. This is again evidenced by the expected visual changes in growth media color from clear yellow to white, and gas production evidenced by movement of the gas exchange membranes. Third, we have completed formal quantitative analysis of the survival of yeast deletion strains dried down on filters by identical techniques to the flight experiment samples. This data shows robust survival of the strains well beyond the storage times in the flight experiment. The data for one of the yeast deletion pools, the homozygous diploid pool, is in press in Applied Microbiology & Biotechnology as R. D'Elia, P.L. Allen, K. Johanson, C.A. Nickerson and T.G. Hammond "Homozygous diploid deletion strains of Saccharomyces cerevisiae that determine lag phase and dehydration tolerance". The data for the other yeast deletion pool in the flight experiment, the heterozygous diploid pool, is in preparation for submission for publication. With regard to the remaining three microorganisms yet to fly (Salmonella typhimurium, Pseudomonas aeruginosa, and Candida albicans), these organisms are currently slated to fly under the experimental payload name "MICROBE" when Shuttle returns to flight and has been selected for flight - Manifest Target: Shuttle flight 12A (launch 12/05). In this regard, the original experiment has been redesigned from a long-term experiment aboard ISS involving continuous culture of the microbes to a short-term experiment due to upmass, volume, and power limitations on upcoming Shuttle flights. These limitations provide considerable constraint to flying the originally proposed MOBIAS culture units, which are each 1 middeck locker and require continuous power. Accordingly, we have redeveloped the experiment to be accomplished in the light weight completely passive GAP (Group Activation Pack/GAP) hardware. Importantly, the PI team has extensive knowledge of the GAP hardware, having successfully flown the GAP hardware on several spaceflight missions (Shuttle and Progress) and most recently with the Yeast-GAP payload. We are currently completing ground-based studies in the GAP hardware in preparation for the MICROBE in-flight experimental payload.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	D'Elia R, Allen PL, Johanson K, Nickerson CA, Hammond TG. "Homozygous diploid deletion strains of Saccharomyces cerevisiae that determine lag phase and dehydration tolerance. " Applied Microbiology and Biotechnology 2005 Jun;67(6):816-26. PMID: 15968571 , Jun-2005
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